

信贷业务流批一体数仓

试运行结果 · 效果图合集

运行环境：Windows 11 | Python 3.14.7 | PySpark 4.2.0 (local[*]) | Java 17.0.10
Kafka 3.9.2 (KRaft 单节点) | Flink 1.20.0 (JobManager + TaskManager 本地集群)
运行时间：2026-09-23 | 代码仓库：github.com/linque0/work

关键数字	
冒烟测试	19 passed, 0 failed (两业务日 2000 客户 / 5000 申请)
流批对账	repay_cnt / repay_amt 相对偏差均 0.0% (容差 1%)
数据质量	27/27 条规则 PASS (9 张表 × 主键/空值率/行数波动)
Schema 漂移	11/11 表 PASS (parquet footer 直比, 不起 Spark)
Kafka→Flink	3557 条消息全量消费 → 1295 个 1min 窗口落 filesystem sink
数据倾斜	95% 流量打 100 热点客户: task 耗时 max 76ms → 51ms (median 60→37ms)
小文件治理	8 个文件 (0.023MB/个) → 1 个 (0.138MB), 原子替换

复现方式

```
$ bash scripts/infra_up.sh

$ bash scripts/run_realtime_demo.sh 2026-09-23 # 复跑加 SKIP_PRODUCE=1

$ python tests/smoke_test.py --n-customers 2000 --n-applications 5000
```

试运行中发现并修复的工程问题

1. 无界流式作业无法判定「消费完成」（sink 停在 694/3557）→ 源表加 scan.bounded.mode=latest-offset，消费到启动位点即 FINISHED、窗口全部触发；
2. curl -o /dev/null 在 MSYS2_ARG_CONV_EXCL 下写 POSIX /dev/null 失败返回非零，集群在线被误判未就绪 → 探测改 shell 重定向；
3. Kafka topic 跨会话累积导致同一 dt 消息翻倍、口径失真（Windows 删 topic 曾触发 broker 退出）→ 重置 topic 后只生产一次；
4. 冒烟测试重跑时同名 landing 文件被 availableNow 视为已处理跳过 → 加测试状态隔离（清 checkpoint / 实时输出表 / landing 批次）；
5. JS 驱动页面（Streamlit / Flink WebUI）用无头浏览器只拍到骨架屏 → 改用 CDP 轮询页面条件后整页截图。

说明：以上问题均在本次试运行中实测复现后修复，修复内容与踩坑记录已写入仓库 README「Kafka/Flink 工程细节」；Mimosa 深度静态扫描 0 发现（依赖分析部分覆盖，未做运行时执行验证）。

信贷业务流批一体数仓 · 试运行结果报告

运行环境: Windows 11 | Python 3.14.7 | PySpark 4.2.0 (local[*]) | Java 17.0.10 | Kafka 3.9.2 (KRaft 单节点) | Flink 1.20.0 (JM+TM 本地集群)
运行时间: 2026-09-23 | 仓库: github.com/linque0/work | 冒烟测试 19/19 PASS | 流批对账 repay_amt 偏差 0.0%

① 端到端冒烟测试 (tests/smoke_test.py)

两个业务日全链路: 生成 → ODS → DWD(SCD2 拉链表) → DWS → ADS → 实时 → 质量 → 对账, 外加 TableStore 快照链与 Schema 校验断言 (2000 客户 / 5000 申请 × 2 日)

```
— gen 2026-09-21 PASS
— ods 2026-09-21 PASS
— dwd 2026-09-21 PASS
— dws 2026-09-21 PASS
— ads 2026-09-21 PASS
— streaming once 2026-09-21 PASS
— quality 2026-09-21 PASS
— reconcile 2026-09-21 PASS
— gen 2026-09-22 PASS
— ods 2026-09-22 PASS
— dwd 2026-09-22 PASS
— dws 2026-09-22 PASS
— ads 2026-09-22 PASS
— streaming once 2026-09-22 PASS
— quality 2026-09-22 PASS
— reconcile 2026-09-22 PASS
[governance] dwd_dim_customer snapshots=11
- snapshot=1790137952951 parent=1790137927202 rows=2097
- snapshot=1790137927202 parent=1790137836596 rows=2097
... （共 11 个快照, 此处省略中间）
- snapshot=1790105515661 parent=1790105491993 rows=82
- snapshot=1790105491993 parent=None rows=80
— schema drift PASS
    2000 ods_customer
    5000 ods_loan_application
    3613 ods_repay_flow
    2097 dwd_dim_customer
    5000 dwd_fact_loan_apply
    3613 dwd_fact_repay
    1984 dws_customer_day
    5000 ads_scorecard_features
SMOKE TEST DONE: 19 passed, 0 failed
ALL PASS
```

信贷业务流批一体数仓 · 试运行结果报告

运行环境: Windows 11 | Python 3.14.7 | PySpark 4.2.0 (local[*]) | Java 17.0.10 | Kafka 3.9.2 (KRaft 单节点) | Flink 1.20.0 (JM+TM 本地集群)
运行时间: 2026-09-23 | 仓库: github.com/linque0/work | 冒烟测试 19/19 PASS | 流批对账 repay_amt 偏差 0.0%

② Kafka + Flink 实时链路全流程（dt=2026-09-23）

等价 Airflow DAG credit_dw_daily 手动序列: gen→ods→dwd→dws→ads→flink_submit→quality→reconcile→compaction→schema→lineage (5000 申请 / 3557 还款事件)

```
##### 0/11 重置 Flink sink 目录（本地演示产物，仓库忽略该路径） #####
##### 1/11 生成业务数据（批量 CSV；默认 一sink kafka 双写消息总线） #####
[demo] SKIP_PRODUCE=1: 跳过 Kafka 生产，只重建批量 CSV
[gen] mode=synthetic dt=2026-09-23 customers=2000 applications=5000 repay_flows=3557
##### 2/11 ODS 贴源层（显式 Schema + 审计字段） #####
[ods] customer dt=2026-09-23 rows=2000 -> <项目根>\warehouse\ods\customer\dt=2026-09-23
[ods] loan_application dt=2026-09-23 rows=5000 -> <项目根>\warehouse\ods\loan_application\dt=2026-09-23
[ods] repay_flow dt=2026-09-23 rows=3557 -> <项目根>\warehouse\ods\repay_flow\dt=2026-09-23
##### 3/11 DWD 明细层（TableStore 拉链表 SCD2 合并） #####
[dwd] dwd_dim_customer snapshot=1790138150695 -> <项目根>\warehouse\tables\dwd_dim_customer\snapshots\1790138150695\data
[dwd] dwd_fact_loan_apply dt=2026-09-23 rows=5000
[dwd] dwd_fact_repay dt=2026-09-23 rows=3557
##### 4/11 DWS 汇总层（按日汇总 + 客户累计） #####
[dws] dws_apply_day dt=2026-09-23 rows=1
[dws] dws_customer_day dt=2026-09-23 rows=2000
##### 5/11 ADS 应用层（风控日报 + 评分卡特征宽表） #####
[ads] ads_risk_daily rows=3
[ads] ads_scorecard_features dt=2026-09-23 rows=5000
[ads] exported dashboard snapshot (history up to dt=2026-09-23)
##### 6/11 提交 Flink SQL 实时作业（双原表双消费者组） #####
[flink] submitting job from C:/flink/submit-job.sql
[flink] job submitted (check http://localhost:8081)
##### 7/11 等待 Flink 1min 窗口数据落 filesystem sink ... #####
[wait] flink sink ready after ~0s (files=1)
##### 8/11 数据质量检查（主键唯一 / 空值率 / 行数波动） #####
[quality] PASS customer rows=2000 (pk_uniqueness=1.0; null_rate=0.0; rowcount_volatility=0.0)
[quality] PASS loan_application rows=5000 (pk_uniqueness=1.0; null_rate=0.0; rowcount_volatility=0.0)
[quality] PASS repay_flow rows=3557 (pk_uniqueness=1.0; null_rate=0.0; rowcount_volatility=0.0155)
[quality] PASS dwd_dim_customer rows=2097 (pk_uniqueness=1.0; null_rate=0.0; rowcount_volatility=0.0)
[quality] PASS dwd_fact_loan_apply rows=5000 (pk_uniqueness=1.0; null_rate=0.0; rowcount_volatility=0.0)
[quality] PASS dwd_fact_repay rows=3557 (pk_uniqueness=1.0; null_rate=0.0; rowcount_volatility=0.0155)
[quality] PASS dws_apply_day rows=1 (pk_uniqueness=1.0; null_rate=0.0; rowcount_volatility=0.0)
[quality] PASS dws_customer_day rows=2000 (pk_uniqueness=1.0; null_rate=0.0; rowcount_volatility=0.0081)
[quality] PASS ads_scorecard_features rows=5000 (pk_uniqueness=1.0; null_rate=0.0; rowcount_volatility=0.0)
[quality] report -> <项目根>\quality\quality_report.json
##### 9/11 流批对账（离线 T+1 vs Flink filesystem sink） #####
[reconcile] PASS repay_cnt: batch=3557 realtime=3557 rel_diff=0.0 (tol=0.01, src=flink-filesystem-sink)
[reconcile] PASS repay_amt: batch=19542249.63 realtime=19542249.63 rel_diff=0.0 (tol=0.01, src=flink-filesystem-sink)
[reconcile] report -> <项目根>\quality\reconcile_report.json
##### 10/11 小文件治理 compaction (dwd_fact_repay) #####
[compaction] dwd_fact_repay dt=2026-09-23: files 1->1, avg 0.136MB->0.136MB
##### 11/11 Schema 漂移校验 + 列级血缘渲染 #####
[schema] PASS ods.customer (11 cols)
[schema] PASS ods.loan_application (17 cols)
[schema] PASS ods.repay_flow (10 cols)
[schema] PASS dwd.dwd_dim_customer (10 cols)
[schema] PASS dwd.dwd_fact_loan_apply (17 cols)
[schema] PASS dwd.dwd_fact_repay (11 cols)
[schema] PASS dws.dws_apply_day (11 cols)
[schema] PASS dws.dws_customer_day (11 cols)
[schema] PASS ads.ads_risk_daily (12 cols)
[schema] PASS ads.ads_scorecard_features (25 cols)
[schema] PASS flink.repay_metrics_fs (json, 1 files)
[schema] report -> <项目根>\quality\schema_drift_report.json
[lineage] written -> <项目根>\docs\lineage.md
##### 附: Kafka topic 现状与消息抽样 #####
##### DONE 2026-09-23 #####
```

信贷业务流批一体数仓 · 试运行结果报告

运行环境: Windows 11 | Python 3.14.7 | PySpark 4.2.0 (local[*]) | Java 17.0.10 | Kafka 3.9.2 (KRaft 单节点) | Flink 1.20.0 (JM+TM 本地集群)

运行时间: 2026-09-23 | 仓库: github.com/linque0/work | 冒烟测试 19/19 PASS | 流批对账 repay_amt 偏差 0.0%

③ 流批对账 (离线 T+1 vs Flink 实时)

同一批还款事件双跑：生成器分别落批量快照与 Kafka topic, Flink 1min 窗口聚合后与离线明细比对

对账通过 | 实时数据源: **flink-filesystem-sink** | 业务日期: **2026-09-23**

指标	离线 T+1	Flink 实时	相对偏差	容差	结论
repay_cnt	3557	3557	0.0	0.01	PASS
repay_amt	19542249.63	19542249.63	0.0	0.01	PASS

信贷业务流批一体数仓 · 试运行结果报告

运行环境: Windows 11 | Python 3.14.7 | PySpark 4.2.0 (local[*]) | Java 17.0.10 | Kafka 3.9.2 (KRaft 单节点) | Flink 1.20.0 (JM+TM 本地集群)
运行时间: 2026-09-23 | 仓库: github.com/linque0/work | 冒烟测试 19/19 PASS | 流批对账 repay_amt 偏差 0.0%

④ 数据质量规则与行数台账 (quality/checks.py)

主键唯一率 / 关键列空值率 / 行数波动三类规则，任一失败即阻断下游 (Airflow 标红)

业务日期 **2026-09-23** | 共 27 条规则，**27/27 PASS**

表.规则	行数	明细	结论
customer.pk_uniqueness	2000	2000/2000	PASS
customer.null_rate	2000		PASS
customer.rowcount_volatility	2000	prev=2000	PASS
loan_application.pk_uniqueness	5000	5000/5000	PASS
loan_application.null_rate	5000		PASS
loan_application.rowcount_volatility	5000	prev=5000	PASS
repay_flow.pk_uniqueness	3557	3557/3557	PASS
repay_flow.null_rate	3557		PASS
repay_flow.rowcount_volatility	3557	prev=3613	PASS
dwd_dim_customer.pk_uniqueness	2097	2097/2097	PASS
dwd_dim_customer.null_rate	2097		PASS
dwd_dim_customer.rowcount_volatility	2097	prev=2097	PASS
dwd_fact_loan_apply.pk_uniqueness	5000	5000/5000	PASS
dwd_fact_loan_apply.null_rate	5000		PASS
dwd_fact_loan_apply.rowcount_volatility	5000	prev=5000	PASS
dwd_fact_repay.pk_uniqueness	3557	3557/3557	PASS
dwd_fact_repay.null_rate	3557		PASS
dwd_fact_repay.rowcount_volatility	3557	prev=3613	PASS
dws_apply_day.pk_uniqueness	1	1/1	PASS
dws_apply_day.null_rate	1		PASS
dws_apply_day.rowcount_volatility	1	prev=1	PASS
dws_customer_day.pk_uniqueness	2000	2000/2000	PASS
dws_customer_day.null_rate	2000		PASS
dws_customer_day.rowcount_volatility	2000	prev=1984	PASS

信贷业务流批一体数仓 · 试运行结果报告

运行环境: Windows 11 | Python 3.14.7 | PySpark 4.2.0 (local[*]) | Java 17.0.10 | Kafka 3.9.2 (KRaft 单节点) | Flink 1.20.0 (JM+TM 本地集群)

运行时间: 2026-09-23 | 仓库: github.com/linque0/work | 冒烟测试 19/19 PASS | 流批对账 repay_amt 偏差 0.0%

⑤ Kafka 总线证据 (topic 描述 + 消息抽样)

生成器 producer 写入 repay_stream (3 分区) , Flink 双消费者组独立消费

```
$ kafka.tools.TopicCommand --describe --topic repay_stream
Topic: repay_stream      TopicId: SpRcb6jhSySMH86F8uI0dw PartitionCount: 3      ReplicationFactor: 1      Configs: segment.bytes=1073741824
    Topic: repay_stream   Partition: 0    Leader: 1      Replicas: 1      Isr: 1 Elr:      LastKnownElr:
    Topic: repay_stream   Partition: 1    Leader: 1      Replicas: 1      Isr: 1 Elr:      LastKnownElr:
    Topic: repay_stream   Partition: 2    Leader: 1      Replicas: 1      Isr: 1 Elr:      LastKnownElr:

$ ConsoleConsumer --topic repay_stream --from-beginning --max-messages 2
{"flow_id": "R2026092300000004", "application_id": "A2026092300000006", "customer_id": "C001416", "event_time": "2026-09-23 14:39:43", "repay_amount": 4054.47, "overdue_days": 0}
{"flow_id": "R2026092300000013", "application_id": "A2026092300000017", "customer_id": "C001727", "event_time": "2026-09-23 22:35:53", "repay_amount": 16697.0, "overdue_days": 0}
```

信贷业务流批一体数仓 · 试运行结果报告

运行环境: Windows 11 | Python 3.14.7 | PySpark 4.2.0 (local[*]) | Java 17.0.10 | Kafka 3.9.2 (KRaft 单节点) | Flink 1.20.0 (JM+TM 本地集群)

运行时间: 2026-09-23 | 仓库: github.com/linque0/work | 冒烟测试 19/19 PASS | 流批对账 repay_amt 偏差 0.0%

⑥ 数据服务 API 响应 (FastAPI :8000)

风控日报 / 维度当前版本 / 评分卡特征点查: parquet 直读 + 60s TTL 缓存, Web 服务不起 Spark

```
$ curl -s http://127.0.0.1:8000/health
{"status":"ok","cache_entries":5}
$ curl -s 'http://127.0.0.1:8000/api/v1/risk/daily?dt=2026-09-23'
{
  "dt": "2026-09-23",
  "data": {
    "apply_date": "2026-09-23",
    "apply_cnt": 5000,
    "approve_cnt": 3557,
    "approve_rate": 0.7114,
    "loan_amt_sum": 544024659.0,
    "avg_interest_rate": 0.099858540000000015,
    "default_cnt": 471,
    "default_rate": 0.1324,
    "apply_cnt_7d": 15000,
    "loan_amt_7d": 1654129367.0,
    "apply_cnt_dod": 0.0,
    "etl_time": "2026-09-23T04:36:47.173164"
  }
}
```

```
$ curl -s http://127.0.0.1:8000/api/v1/dim/customer/C000001
{
  "data": {
    "customer_id": "C000001",
    "start_date": "2026-09-21",
    "gender": "F",
    "age": 34,
    "education": "本科",
    "marital_status": "离异",
    "occupation": "其他",
    "city": "北京",
    "customer_sk": -4445154481378003693,
    "end_date": "9999-12-31"
  }
}
```

信贷业务流批一体数仓 · 试运行结果报告

运行环境: Windows 11 | Python 3.14.7 | PySpark 4.2.0 (local[*]) | Java 17.0.10 | Kafka 3.9.2 (KRaft 单节点) | Flink 1.20.0 (JM+TM 本地集群)
运行时间: 2026-09-23 | 仓库: github.com/linque0/work | 冒烟测试 19/19 PASS | 流批对账 repay_amt 偏差 0.0%

⑦ 数据治理（Schema 漂移 + 快照链 + 倾斜基准）

parquet footer 直比可进 CI; TableStore 不可变快照链支撑时间旅行; 倾斜治理用加盐两阶段聚合

Schema 漂移检测 (parquet footer 直比, 不起 Spark) : **11/11 PASS**

表	注册列/实际列	漂移	结论
ods.customer	11 / 11	无漂移	PASS
ods.loan_application	17 / 17	无漂移	PASS
ods.repay_flow	10 / 10	无漂移	PASS
dwd.dwd_dim_customer	10 / 10	无漂移	PASS
dwd.dwd_fact_loan_apply	17 / 17	无漂移	PASS
dwd.dwd_fact_repay	11 / 11	无漂移	PASS
dws.dws_apply_day	11 / 11	无漂移	PASS
dws.dws_customer_day	11 / 11	无漂移	PASS
ads.ads_risk_daily	12 / 12	无漂移	PASS
ads.ads_scorecard_features	25 / 25	无漂移	PASS
flink.repay_metrics_fs	- / -	无漂移	PASS

```
snapshot=1790138150695 parent=1790137952951 2026-09-23T12:35:55 merge_upsert rows=2097 keys=[ customer_id, start_date ]
snapshot=1790137952951 parent=1790137927202 2026-09-23T12:32:33 merge_upsert rows=2097 keys=[ customer_id, start_date ]
snapshot=1790137927202 parent=1790137836596 2026-09-23T12:32:09 merge_upsert rows=2097 keys=[ customer_id, start_date ]
snapshot=1790137836596 parent=1790137798328 2026-09-23T12:30:37 merge_upsert rows=2080 keys=[ customer_id, start_date ]
snapshot=1790137798328 parent=1790137696407 2026-09-23T12:29:59 merge_upsert rows=2044 keys=[ customer_id, start_date ]
snapshot=1790137696407 parent=1790135706024 2026-09-23T12:28:18 merge_upsert rows=2044 keys=[ customer_id, start_date ]
snapshot=1790135706024 parent=1790135400507 2026-09-23T11:55:10 merge_upsert rows=2002 keys=[ customer_id, start_date ]
snapshot=1790135400507 parent=1790135382911 2026-09-23T11:50:01 merge_upsert rows=82 keys=[ customer_id, start_date ]
snapshot=1790135382911 parent=1790106002918 2026-09-23T11:49:44 merge_upsert rows=82 keys=[ customer_id, start_date ]
snapshot=1790106002918 parent=1790105515661 2026-09-23T03:40:06 merge_upsert rows=82 keys=[ customer_id, start_date ]
snapshot=1790105515661 parent=1790105491993 2026-09-23T03:31:56 merge_upsert rows=82 keys=[ customer_id, start_date ]
snapshot=1790105491993 parent=None 2026-09-23T03:31:32 merge_upsert rows=80 keys=[ customer_id, start_date ]
```

数据倾斜基准 (300,000 行 / 14309 客户 / 16 盐, 95% 流量集中 100 热点客户, Spark EventLog 采集 task 级指标)

方案	task 耗时 max(ms)	task 耗时 median(ms)	shuffle 写(MB)	端到端(ms)
直接 groupBy (倾斜)	76	60	0.11	208
加盐两阶段聚合	51	37	0.11	188

⑧ 小文件治理：8 个小文件（0.023MB/个）→ compaction 原子替换合并为 1 个

信贷业务流批一体数仓 · 试运行结果报告

运行环境: Windows 11 | Python 3.14.7 | PySpark 4.2.0 (local[*]) | Java 17.0.10 | Kafka 3.9.2 (KRaft 单节点) | Flink 1.20.0 (JM+TM 本地集群)

运行时间: 2026-09-23 | 仓库: github.com/linque0/work | 冒烟测试 19/19 PASS | 流批对账 repay_amt 偏差 0.0%

⑧ 小文件治理（构造小文件场景 → compaction 合并）

为演示治理链路：关闭 AQE 分区合并、以 8 个 shuffle 分区重写当日分区制造小文件，再跑 compaction 原子替换合并

表 **dwd_fact_repay** (dt=2026-09-23) | 阈值: 文件数 > 50 或 平均大小 < 8.0MB | 治理已触发

指标	治理前	治理后
文件数	8	1
总大小(MB)	0.181	0.138
平均文件大小(MB)	0.023	0.138

```
##### 小文件治理演示 (dwd_fact_repay dt=2026-09-23) #####
>>> 步骤 1/2: 显式 repartition(8) 重写当日分区, 制造 8 个小文件
WARNING: Using incubator modules: jdk.incubator.vector
[Stage 7:>                                (0 + 8) / 8]
[mk-small-files] rows=3557 rewritten into 8 files -> warehouse/dwd/dwd_fact_repay/dt=2026-09-23
>>> 步骤 2/2: compaction 合并 (阈值触发 + 原子替换)
错误: 无法终止 PID 62944 (属于 PID 60512 子进程)的进程。
错误: 无法终止 PID 60512 (属于 PID 67520 子进程)的进程。
错误: 无法终止 PID 67520 (属于 PID 39684 子进程)的进程。
WARNING: Using incubator modules: jdk.incubator.vector
[Stage 1:>                                (0 + 1) / 1]
[compaction] dwd_fact_repay dt=2026-09-23: files 8->1, avg 0.023MB->0.138MB
##### DONE #####
错误: 无法终止 PID 12164 (属于 PID 67212 子进程)的进程。
成功: 已终止 PID 67212 (属于 PID 22396 子进程)的进程。
成功: 已终止 PID 22396 (属于 PID 28848 子进程)的进程。
```

- Apache Flink Dashboard
- Overview
- Jobs
- Running Jobs
- Completed Jobs
- Task Managers
- Job Manager
- Submit New Job



Available Task Slots

4

Total Task Slots 4 | Task Managers 1

Running Jobs

0

Finished 6 | Canceled 10 | Failed 0

Running Job List

Job Name	Start Time	Duration	End Time	Tasks	Status
----------	------------	----------	----------	-------	--------



No Data

Completed Job List

Job Name	Start Time	Duration	End Time	Tasks	Status
insert-into_default_catalog.default_database.repay_metrics_kafka	2026-09-23 12:16:09.085	5m 5s 346ms	2026-09-23 12:21:14.431	22	CANCELED
insert-into_default_catalog.default_database.repay_metrics_kafka	2026-09-23 12:21:43.861	163ms	2026-09-23 12:21:44.024	22	FINISHED
insert-into_default_catalog.default_database.repay_metrics_kafka	2026-09-23 12:36:58.558	148ms	2026-09-23 12:36:58.706	22	FINISHED
insert-into_default_catalog.default_database.repay_metrics_fs	2026-09-23 12:21:43.414	183ms	2026-09-23 12:21:43.597	22	FINISHED
insert-into_default_catalog.default_database.repay_metrics_fs	2026-09-23 12:36:58.164	149ms	2026-09-23 12:36:58.313	22	FINISHED
insert-into_default_catalog.default_database.repay_metrics_kafka	2026-09-23 02:51:40.475	8h 56m 57s	2026-09-23 11:48:37.902	22	CANCELED
insert-into_default_catalog.default_database.repay_metrics_fs	2026-09-23 12:16:08.601	5m 6s 90ms	2026-09-23 12:21:14.691	22	CANCELED
insert-into_default_catalog.default_database.repay_metrics_fs	2026-09-23 12:27:28.777	185ms	2026-09-23 12:27:28.962	22	FINISHED
insert-into_default_catalog.default_database.repay_metrics_kafka	2026-09-23 12:27:29.156	154ms	2026-09-23 12:27:29.310	22	FINISHED
insert-into_default_catalog.default_database.repay_metrics_fs	2026-09-23 02:51:40.029	8h 56m 57s	2026-09-23 11:48:37.662	22	CANCELED

信贷风控日报看板

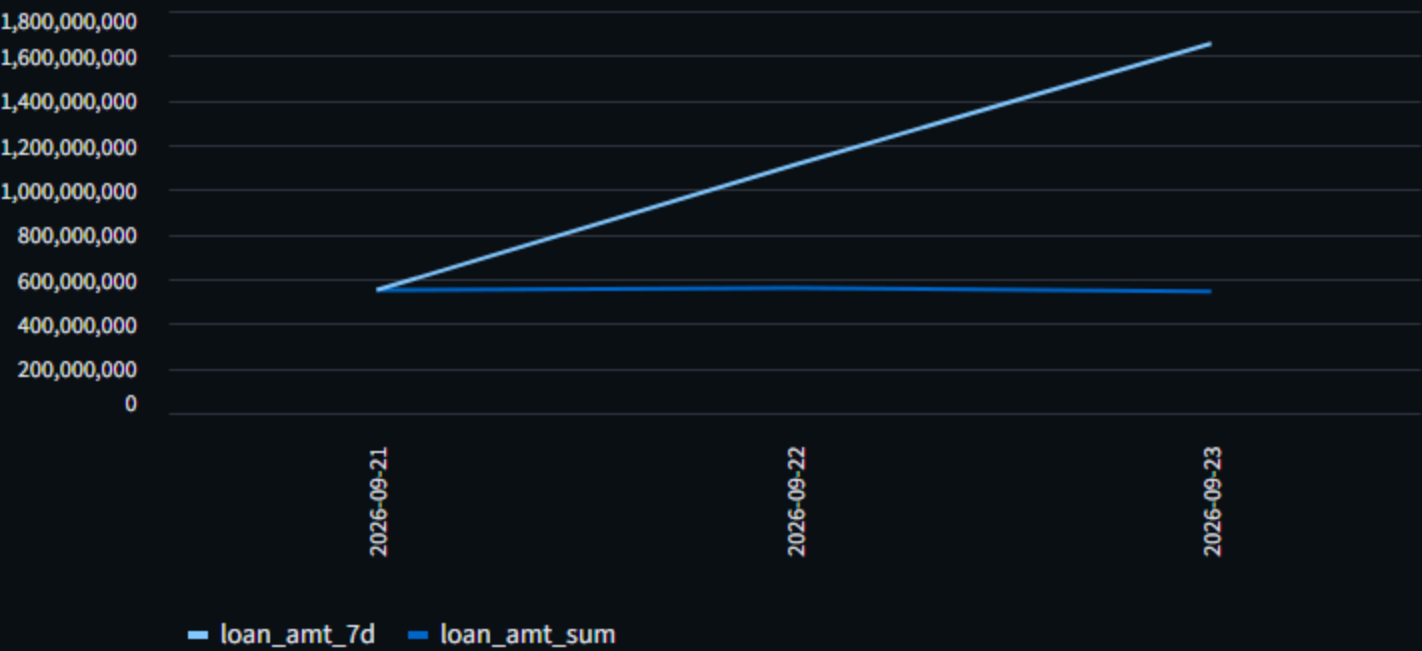
数据来源：ADS 层 ads_risk_daily（离线 T+1，流批对账保障一致性）

业务日期	当日申请量	通过率	当日放款金额	违约率
2026-09-23	5,000	71.1%	544,024,659	13.2%

申请量 / 通过量趋势



放款金额与 7 日滚动



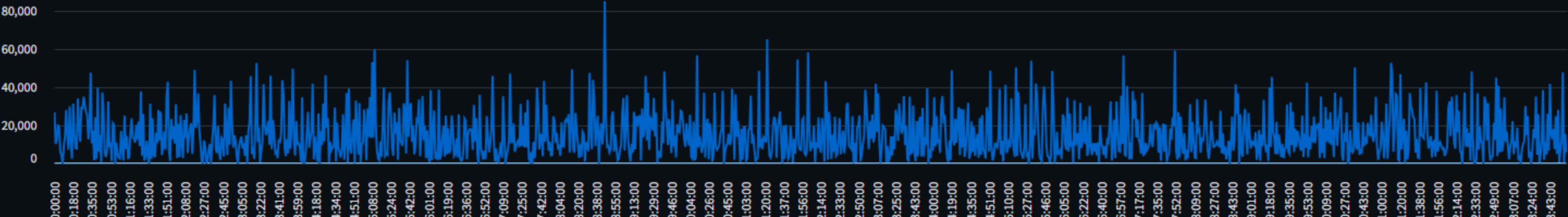
风控日报明细

apply_date	apply_cnt	approve_cnt	approve_rate	loan_amt_sum	avg_interest_rate	default_cnt	default_rate	apply_cnt_7d	apply_cnt_dod
2026-09-21	5000	3607	0.7214	550245192	0.0994	532	0.1475	5000	None
2026-09-22	5000	3613	0.7226	559859516	0.1	475	0.1315	10000	0
2026-09-23	5000	3557	0.7114	544024659	0.0999	471	0.1324	15000	0

实时还款指标（Flink，1 分钟窗口）

最新业务日	实时消息数	实时还款金额
2026-09-23	3,557	19,542,250

流批对账保障：同一指标离线 T+1 与本实时链路偏差 < 1%（quality/reconcile.py）



信贷数仓数据服务

1.0.0

OAS 3.1

/openapi.json

default



GET	/health	Health	▼
GET	/api/v1/risk/daily	Risk Daily	▼
GET	/api/v1/scorecard/features/{application_id}	Scorecard Features	▼
GET	/api/v1/customer/{customer_id}/features	Customer Features	▼
GET	/api/v1/dim/customer/{customer_id}	Dim Customer	▼

Schemas



HTTPValidationError	>	Expand all	object
ValidationError	>	Expand all	object